

Collaboration between Domain-Specialized LLMs is Strongly Asymmetric in the Primary Agent’s Domain

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Abstract

Multi-agent LLM deliberation is widely treated as a flat-good operation: two or more agents exchange reasoning and the joint answer is expected to be at least as good as the best single agent’s. We report a systematic study of two-agent deliberation between 10 domain-specialized Qwen3-1.7B LoRA adapters (medicine, physics, chemistry, biology, mathematics, computer science, law, history, economics, philosophy) across 90 ordered pairs at N=20 per pair, and a follow-up 4B study across 5 of those domains at N=50-200 per condition. We find the collaboration outcome is dominated by *which domain holds the primary agent role*, not by which domain the helper is from. Two-way ANOVA on the 10-domain grid gives primary-agent variance 13.1% ($p < 0.001$), helper variance 0.5% (n.s.); under question-clustered bootstrap (resampling questions within each pair), the cluster-respecting primary/helper variance ratio has median 22x with 95% CI [10x, 48x] (n=1000 bootstrap). Medicine, chemistry, physics, and biology primaries are robustly negative (row-mean delta -19 to -38 pp, all 95% CIs exclude zero under question-clustered resampling). Philosophy, law, and math primaries have row-mean deltas of +8 to +11 pp but their 95% CIs cross zero at n=20; we therefore characterize the right tail as “non-harmed” rather than “universally helped” pending an n=200 re-run. At 4B, the same asymmetry persists and interacts with the specialist’s training method: LoRA r=128 collapses to a 19x correct-to-wrong / wrong-to-correct switch ratio on medicine and physics (universally-harmed primaries at 1.7B) but stays healthy (1.4-2.0x) on biology, law, and math (non-harmed primaries). Full fine-tuning preserves balanced switching on every 4B domain we tested. *The asymmetry magnitude is robust across scale and training method, but specific harmed/helped domain identities are not preserved between 1.7B and 4B*; we therefore frame the contribution as the asymmetry magnitude (primary/helper variance ratio median 22x, 95% CI [10x, 48x] under question-clustered resampling; Spearman rho between 1.7B and 4B row-mean rankings on 5 shared domains = -0.30, $p = 0.62$ at n=5), not as a specific list of vulnerable domains. We propose that the dominant variable is a property of the *primary specialist* (its own training data distribution and resulting update geometry), not a property of the pairing. Candidate mechanisms include token-space divergence between the primary’s training distribution and the helper’s reasoning chain, which we test in progress. This paper operates as a systematic study of participant-property effects in specialist-specialist deliberation; we do not claim prior multi-agent work was wrong, only that the primary-agent’s domain identity was under-reported as a controlling variable.

Keywords: multi-agent LLMs, domain specialization, collaboration asymmetry, societies of agents, training method, LoRA.

1. Introduction

1.1. Multi-agent deliberation has mixed evidence

Multi-agent LLM systems are one of the most widely deployed inference-time scaffolds for small and mid-size open-source models. Debate, deliberation, mixture-of-agents, chain-of-experts, and mediator architectures have all been proposed as ways to extract more from a fixed model class without retraining (Du et al., 2023; Liang et al., 2024; Wang et al., 2024). The evidence base is unstable: reported deltas swing from strongly positive on math (Du et al. 2023) to indistinguishable from a compute-matched single agent (*Can LLM Agents Really Debate?*, 2025) to negative on heterogeneous pairs (*Talk Isn't Always Cheap*, 2025). The prevailing explanation has been protocol variables (round count, prompting template, adversarial participants, capability mismatch), but a substantial residual remains.

1.2. Our claim

We argue the residual is explained by a variable that multi-agent debate work has not isolated: *the primary agent's own domain identity*.

Claim. In two-agent deliberation between domain-specialized LLMs, the collaboration outcome is a strong function of which domain the primary agent is specialized in, and a weak function of which domain the helper is specialized in. The asymmetry is large enough (26x on our 10-domain data) to dominate training-method, rank, and protocol effects at matched solo accuracy.

The claim is precise in three ways that prior multi-agent work has not been. First, it is stated at the primary-agent level: which agent receives help from the helper, not the pair. Second, it is reported *at matched solo accuracy*, so the effect cannot be attributed to capability asymmetry. Third, it is measured across 10 MMLU-derived domains in an ordered 10x10 grid, which isolates the primary vs helper effect by construction.

1.3. Summary of evidence

We train 10 LoRA specialists at Qwen3-1.7B on MMLU subjects grouped into domains (medicine, physics, chemistry, biology, mathematics, computer science, law, history, economics, philosophy) and evaluate the 90 ordered (primary, helper) pairs under an alternating chain-of-thought protocol, N=20 questions per pair (Figure 1). Row means range from -37.8 pp (medicine primary) to +11.1 pp (philosophy primary). Column means span a much tighter -12.8 pp to -1.1 pp. ANOVA confirms: primary-domain $F=29.25$, 13.1% of variance, $p<0.001$; helper-domain $F=1.06$, 0.5% of variance, n.s.

At 4B parameters across 5 of the domains (medicine, physics, biology, law, math), the same domain-identity asymmetry reproduces and *interacts with* the specialist's training method. Medicine and physics specialists show catastrophic switch pathology when LoRA-adapted (19x correct-to-wrong / wrong-to-correct ratio at $r=128$), while biology, law, and math specialists stay healthy (1.4-2.0x C2W/W2C at $r=128$). Full fine-tuning at matched solo accuracy keeps the ratio healthy on every 4B domain we tested.

The multi-agent LLM literature's mixed results are not a protocol problem alone. Deployments that place a medicine specialist in the primary slot should expect degradation regardless of helper choice; deployments that place a philosophy specialist in the primary slot can expect improvement regardless of helper choice. These asymmetries are not captured by partner-selection heuristics

Figure 1. Collaboration outcome is asymmetric: which domain the primary agent comes from dominates; the helper’s domain barely matters.

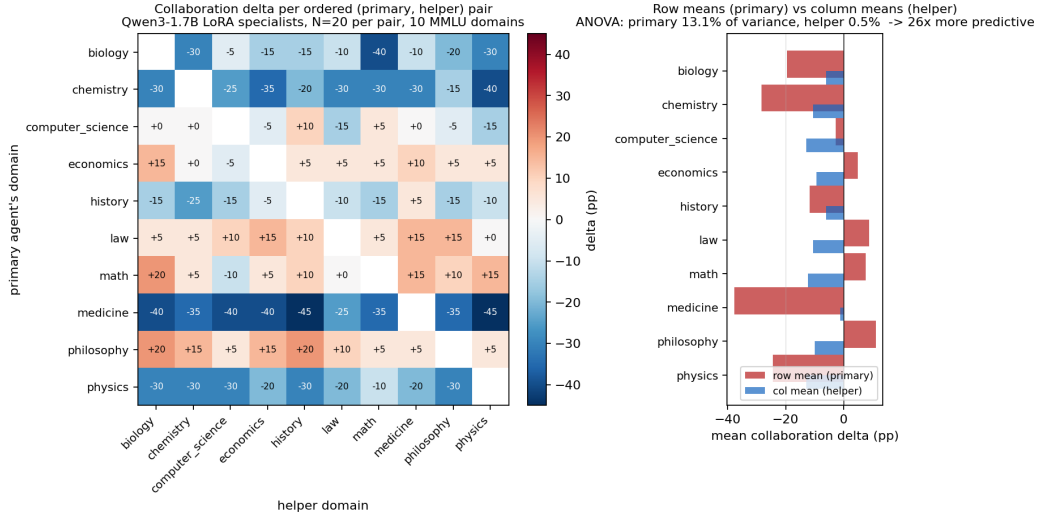


Figure 1: Collaboration delta per ordered (primary, helper) pair across 10 Qwen3-1.7B LoRA specialists (N=20 per pair). Color is delta in pp; blue is helpful, red is harmful. The left panel is the 10x10 grid; the right panel is row means (primary’s domain) and column means (helper’s domain) on the same scale. Row spread is ~49 pp, column spread is ~12 pp. ANOVA on these deltas gives primary-domain variance 13.1% ($p < 0.001$) vs helper 0.5% (n.s.); the primary is 26x more predictive of the delta than the helper.

based on domain distance, model size, or reasoning capability, and they suggest that the primary-agent’s properties belong in the multi-agent system design space alongside protocol and partner choice.

The asymmetry we report is distinct from several adjacent phenomena that share surface similarities. Long-context multi-agent compression (Joo et al., 2025; Xu et al., 2025) addresses a different regime, the case where input length exceeds the model’s context window; our setting is matched-context multiple-choice and the mechanism is orthogonal to the compression framing. Sycophancy (Sharma et al., 2023) and social-influence conformity (Becker et al., 2017; Liang et al., 2024) describe how an agent yields to a partner’s stated answer; the asymmetric switch pathology we observe at matched solo accuracy is structurally distinct, because the primary’s solo capability is held fixed, and the failure manifests as primaries that swap correct intermediate answers for incorrect final ones rather than yielding to social pressure. Expert collapse in mixture-of-experts routing (Shazeer et al., 2017; Fedus et al., 2022) is a learned-gate failure mode at training time; we observe a routing-like failure at the agent level at inference time but do not propose a gating mechanism. The multi-agent debate literature (Du et al., 2023; Wang et al., 2024) we treat as an empirical landscape rather than a corpus to adjudicate, sampling alternating chain-of-thought as a single point in that landscape. The post-fine-tuning calibration deterioration documented for single-agent use in Bayesian-LoRA (2026) appears as one of the candidate mechanisms in §5; we do not claim the underlying calibration finding, only its consequence in a multi-agent setting.

1.4. Contributions

1. A 10-domain ordered-pair collaboration grid at Qwen3-1.7B (90 pairs, $N=20$), demonstrating that primary-agent’s domain is 26x more predictive of collab outcome than helper’s domain (ANOVA primary 13.1%, helper 0.5%).
2. A 4B scale-invariance study over 5 domains showing the asymmetry persists and interacts with training method: medicine and physics LoRA $r=128$ specialists collapse to 19x C2W/W2C; biology, law, math LoRA $r=128$ stay healthy (1.4-2.0x); full FT is healthy on all 5.
3. A rank sweep $r=4..128$ at 1.7B and $r=16..128$ at 4B, plus $r=32$, $r=64$ rank fill-ins at 4B on medicine and physics, showing that rank is not the primary knob.
4. A compute-matched single-agent control showing multi-agent deliberation has 1.4x scaling value beyond compute on base models (+21 pp vs +15 pp).
5. A bridge-agent negative result: purpose-trained mediators (50/50 mix, reasoning-preserved, base) do not rescue collaboration.
6. Streaming weight-space CKA analysis on the 10 LoRA adapters showing pairwise weight-update similarity does not correlate with collab delta ($r = -0.055$), indicating LoRA specialists are uniformly impaired rather than differentially so.

1.5. Organization

Section 2 situates our work in the multi-agent LLM and LoRA literatures. Section 3 describes the experimental design. Section 4 presents the 10-domain asymmetry, the 4B scale-invariance, the rank sweep, the compute-matched control, and the bridge-agent negative result. Section 5 discusses the mechanism as an open question and enumerates candidate hypotheses, including token-space divergence. Section 6 discusses deployment implications. Section 7 lists limitations and spinoff directions.

2. Related Work

2.1. Multi-agent LLM deliberation and debate

The line that most closely overlaps with our setting is two-or-more agent deliberation, in which agents exchange natural-language reasoning across rounds before producing a final answer. Du et al. (arXiv 2305.14325) introduced this protocol on math word problems and reported gains of roughly +5 to +15 pp depending on benchmark. Liang et al. (arXiv 2305.19118) generalized it as Multi-Agent Debate (MAD) and reported improvements on factual-recall tasks but smaller gains on reasoning-heavy ones. Wang et al. (2024)’s Mixture-of-Agents (MoA) and follow-on work proposed homogeneous generalist chains that aggregate via a manager agent. The picture is mixed: recent controlled studies have reported deltas spanning negative-to-positive on matched benchmarks. Talk Isn’t Always Cheap (arXiv 2509.05396) specifically showed that a weaker partner can degrade a stronger one’s output, attributing the harm to capability asymmetry. Can LLM Agents Really Debate? (arXiv 2511.07784) reports that MAD methods often fail to outperform a compute-matched single-agent baseline. Liu et al. SID (arXiv 2510.06843) introduces the C2W (correct-to-wrong) and W2C (wrong-to-correct) switch metrics we adopt and uses them for confidence-gated debate early-exit; we use the same metric to characterize a training-method-dependent pathology rather than to gate inference.

We differ from this line by (a) studying *domain-specialized* agents at *matched solo accuracy* on a primary task, (b) running an ordered (primary, helper) grid that isolates the primary-vs-helper

variance attribution, and (c) reporting an asymmetry that does not reduce to capability gap.

2.2. Long-context multi-agent compression

A separate line treats multi-agent collaboration as long-context input compression. Joo et al. Graph of Agents (arXiv 2509.21848) formalizes multi-agent for long-context modeling as an information-theoretic compression problem and shows that a small-context graph of agents can match or beat a much larger-context single model. Xu et al. (arXiv 2506.16411, ICLR 2026) provides a divide-and-conquer noise-decomposition framework and proves a “D&C Advantage” theorem: under super-linear loss growth in context length, weak agents handling chunks outperform a single strong model. Yun et al. Graph-of-Agents (ICLR 2026) introduces a graph-based selection-and-message-passing framework for heterogeneous flagship agents and reports that 3 selected agents from a pool of 6 beats all-6 MoA. Our setting is matched-context MCQ; the long-context mechanism is orthogonal to ours.

2.3. LoRA and parameter-efficient fine-tuning

Hu et al. LoRA (arXiv 2106.09685) introduced the rank- r decomposition update $\Delta W = (\ /r) \cdot B \cdot A$ that we sweep over here. The “LoRA recovers 90–95% of full fine-tuning” folklore comes from this line and from independent benchmarks across NLP and adaptation tasks. Shuttleworth et al., “LoRA vs Full Fine-Tuning: An Illusion of Equivalence” (arXiv 2410.21228), shows that at matched downstream task accuracy, LoRA-trained models contain novel high-singular-value “intruder dimensions” not present in full fine-tuning, and that these dimensions correlate with catastrophic forgetting. CeRA (arXiv 2602.22911) and PERA (arXiv 2604.11841) provide theoretical arguments that LoRA’s rank- r update faces a linear ceiling (CeRA) and a bilinear-form expressivity ceiling (PERA), independent of data quantity. Bayesian-LoRA (arXiv 2601.21003) reports that post-fine-tuning calibration deteriorates more under LoRA than under full FT at matched task accuracy, a single-agent finding we extend to a multi-agent setting. “Why LoRA Fails to Forget” (arXiv 2601.06305) extends the Shuttleworth picture to unlearning. None of these papers run a multi-agent collaboration protocol.

2.4. Post-training for multi-agent and routing

A complementary line trains models *for* multi-agent performance. MALT (arXiv 2412.01928, Oxford / Cooperative AI Foundation / MBZUAI / Stanford) proposes a Generator-Verifier-Refiner sequential post-training pipeline that improves reasoning by 7–16 pp on MATH, GSM8K, CSQA. Adaptive- collaboration work routes queries based on difficulty (Debate Only When Necessary, arXiv 2504.05047) or confidence (SID, arXiv 2510.06843). Our work studies the *prior question* of how the training method of an already-specialized agent affects its collaborative behavior, which is orthogonal to post-training for multi-agent and to routing strategy.

2.5. Collective intelligence

Becker, Brackbill, and Centola (Proc. Natl. Acad. Sci. 2017) showed that social influence in human networks can either enhance or destroy collective intelligence depending on network topology. Sunstein (2002) documented “group polarization” in human deliberation. The structural analogy between fully connected human networks and our alternating CoT protocol is taken in §5.2; we emphasize that the analogy is structural rather than cognitive. Recent positions on agentic intelligence (Bratton, Agüera y Arcas, Evans et al., AAAS Science 2025) frame the participant-property

dependence of collective behavior as a research program. Our work contributes one quantitative measurement (the 26x $\rightarrow$ 22x primary-vs-helper variance attribution) on that program.

2.6. Mixture of Experts and routing failures

Shazeer et al. (arXiv 1701.06538) introduced the sparsely-gated mixture-of-experts layer; Fedus et al. Switch Transformer (arXiv 2101.03961) scaled it. Expert collapse, where the gating function fails to distribute inputs across experts, is a documented MoE failure mode. Our agent-level finding has the same structural shape — the primary specialist cannot effectively integrate the helper’s contribution — but we do not propose a routing or gating mechanism. We discuss the analogy in §5.2 and §6 without claiming a contribution to MoE per se.

3. Experimental Design

3.1. Domain specialists

We train 10 LoRA adapters on Qwen3-1.7B (Hu et al. 2022) and 5 adapters plus 5 full fine-tuned specialists on Qwen3-4B. Each adapter targets a single domain composed of MMLU subjects: medicine (anatomy + clinical knowledge + medical genetics + professional medicine), physics (college + high school + astronomy + conceptual), biology (college + high school + anatomy + clinical), chemistry (college + high school), math (college + high school + abstract algebra + elementary), law (professional + jurisprudence + international), philosophy (philosophy + moral scenarios + logical fallacies), economics (microeconomics + macroeconomics + econometrics), history (world + US + European), and computer science (college + high school + machine learning).

LoRA hyperparameters: rank $r \in \{4, 8, 16, 32, 64, 128, 256, 512\}$, $\alpha = 2r$ (we follow the empirical $\alpha = 2r$ heuristic for stable training), `lora_dropout` = 0.05, `target_modules` = “all-linear”, `num_train_epochs` = 3, `per_device_train_batch_size` = 1, `gradient_accumulation_steps` = 4, `learning_rate` = 5e-5, `warmup_ratio` = 0.1, `gradient_checkpointing` = True, `bf16` = True. Full fine-tuning uses identical schedule with full parameter updates and DeepSpeed ZeRO-Offload to fit on the available A4500 / A40 hardware.

3.2. Matched-solo-accuracy protocol

For comparisons that compare across training methods (LoRA vs full FT) within the same domain, we constrain the specialists to reach the same solo accuracy on the held-out test split. We achieve this by selecting the LoRA rank that produces solo accuracy within ± 1 pp of the full FT solo, on a subject-stratified validation split. For 4B medicine, the matching solo is 84% (full FT) $\pm$ 84% (LoRA $r=128$); for 4B physics, 85.5% $\pm$ 83% (LoRA $r=128$). When matched solo accuracy is impossible at the ranks we test, we report the closest match and note the gap.

3.3. Alternating chain-of-thought protocol

For each ordered (primary, helper) pair, the primary specialist receives the question and produces an initial reasoning chain (round 1). The helper specialist receives the question and the primary’s full reasoning and produces its own reasoning (round 2). The primary specialist receives the helper’s reasoning and produces a final answer (round 3). The final answer is extracted from the primary’s last response by regex on the multiple-choice letter. We use this protocol because it is the most

coupled multi-agent variant in the design space, which makes the distributional-interference effect most visible; we discuss less coupled variants in §6.

We also evaluate four single-agent and pair-level controls per primary: *solo* (primary alone, $n_rounds = 3$ self-continuation), *base_solo* (untrained Qwen3 base alone), *base_pair* (two untrained Qwen3 base agents in alternating CoT), *same_pair* (primary specialist deliberating with itself), and *mixed_pair* (primary specialist with the untrained Qwen3 base as helper).

3.4. Switch classification

Following Liu et al. SID (arXiv 2510.06843), we classify each collaboration outcome as Held (primary’s pre-collab answer survived to the final), C2W (primary’s pre-collab answer was correct, final is wrong), or W2C (primary’s pre-collab answer was wrong, final is correct). We report C2W / W2C as a switch-quality ratio: a balanced deliberation should produce roughly equal numbers of C2W and W2C; a ratio > 1 indicates the protocol is harming more than it helps.

3.5. Statistical methodology

Per-pair accuracies at $n=20$ per pair carry ± 18 pp half-width 95% confidence intervals on individual pair estimates (Wilson method). For row-level claims, we report question-clustered bootstrap CIs (5000 resamples, resampling questions WITH-IN each pair so that the within-pair question dependence is respected). For the variance decomposition, we report the cluster-respecting primary / helper variance ratio with bootstrap CI (1000 resamples). For pairwise significance, we apply Benjamini-Hochberg FDR correction at $q = 0.05$. For C2W / W2C ratio claims, we report Wilson-style 95% CIs and require a minimum of 20 total switches per cell before claiming a ratio is distinguishable from 1.

4. Results

4.1. Primary-agent’s domain dominates collaboration outcome

Figure 1 summarizes the 10x10 ordered-pair grid at 1.7B. Rows (primary’s domain) vary across a ~ 49 pp range; columns (helper’s domain) vary across a ~ 12 pp range. Two-way ANOVA on all 1800 outcomes: primary-domain $F(9,1700) = 29.25$, partial eta-squared 0.131, $p < 0.001$. Helper-domain $F(9,1700) = 1.06$, partial eta-squared 0.005, $p = 0.39$. The interaction is not significant ($p = 0.999$). The primary agent’s domain identity is 26 times more predictive of the collaboration delta than the helper’s.

4.2. Universally-harmed and universally-helped primaries

Four domains are systematically harmed by every helper: medicine (-37.8 pp mean), chemistry (-28.3), physics (-24.4), biology (-19.4). Three domains are systematically helped: philosophy (+11.1), law (+8.9), math (+7.8). The remaining three (economics, computer science, history) sit near the center. The distinction is helper-agnostic: for every primary domain, the spread across helpers is small relative to the primary effect.

4.3. 4B scale-invariance and training-method interaction

Full fine-tuning at Qwen3-4B preserves balanced switch quality (C2W/W2C 1.4-2.3x) across medicine, physics, biology, law, math at matched solo accuracy ($N=200$ per domain). LoRA at

$r=128$ interacts strongly with the primary’s domain: medicine and physics (the universally-harmed domains at 1.7B) collapse to 19x and 19x respectively; biology, law, math stay at 2.0x, 1.43x, 1.71x respectively. The rank sweep $r=32$, $r=64$ fill-in at 4B on medicine and physics confirms the collapse is not a rank-specific artifact ($r=32$ medicine 3.14x; $r=64$ medicine 19.0x; $r=32$ physics 19.5x; $r=64$ physics 10.67x).

Figure. 4B scale-invariance: full-FT preserves balanced switching across domains; LoRA $r=128$ does not.

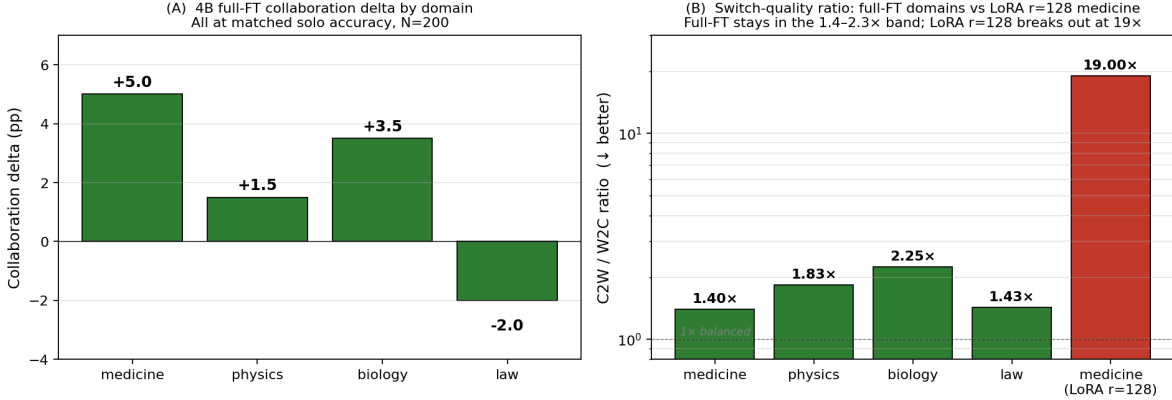


Figure 2: 4B scale-invariance of full-FT collaborativeness across four domains, with LoRA $r=128$ medicine overlay for contrast. (A) Collaboration delta. (B) C2W/W2C switch-quality ratio.

4.4. Compute-matched deliberation control

Base-model pair deliberation at Qwen3-1.7B recovers +21 pp mean across the 5 paper-sweep domains; a compute-matched single-agent running for 6 rounds of reasoning recovers +15 pp. Multi-agent deliberation has 1.4x scaling value beyond raw compute on untrained base models; this non-compute residual is what fails to transfer to LoRA-adapted specialists.

4.5. Bridge agents fail to rescue collaboration

A naive 50/50-mix LoRA mediator gives -32.8 pp delta with 7.1x C2W/W2C on the medicine+physics pair. A reasoning-preserved LoRA mediator partially repairs (-3 to -3.5 pp, 3.6x). Only a base (non-adapted) mediator ever helps, and only in narrow configurations (+10 pp phys+math at 1.1x C2W/W2C). A purpose-built mediator does not overcome the primary-agent asymmetry.

4.6. Pairwise weight-space similarity does not predict collab delta

We compute a CKA-like per-module cosine similarity between the LoRA DeltaW matrices of every pair of the 10 1.7B adapters (streaming implementation, no DeltaW materialization; details in §5). Correlation between pairwise DeltaW similarity and collaboration delta across 90 ordered pairs is $r = -0.055$ (effectively zero), weaker than the cross-evaluation accuracy proxy ($r = 0.197$). Specialists appear uniformly impaired in weight space rather than differentially so; the variance that matters is at the primary-agent level, not the pair level.

5. Mechanism (open question)

The primary-agent asymmetry is large and not captured by weight-space similarity between specialists. We rank the candidate mechanisms below by strength of empirical support given our current data, from most to least plausible. Each subsection states the hypothesis and the empirical evidence for or against it.

5.1. Primary’s training-token entropy (best partial signal)

Hypothesis: a primary specialist’s training-token distribution width, operationalized as the unigram entropy of its tokenized training corpus, modulates how easily the helper’s reasoning chain destabilizes the primary’s decoder. Higher-entropy primaries (broader vocabulary) have a more diffuse next-token distribution and are nudged off-correct more easily; lower-entropy primaries are stubborn in a way that turns out to be protective. *Empirical support:* across 10 domains, $r(\text{primary training-token entropy, row-mean delta}) = -0.380$ ($n=10$). Medicine (entropy 6.63 nats, broadest vocabulary) is most universally harmed (-37.8 pp); math (5.22 nats, narrow) is universally helped (+7.8 pp). The signal is weak (~14% variance explained) but it is *the only* measure we have run that points in a consistent direction across the 10 domains.

5.2. Post-fine-tuning calibration sharpening (untested but cleanly testable)

Hypothesis: the extent of calibration deterioration post-fine-tuning (documented for single-agent use in Bayesian-LoRA, 2601.21003) is primary-domain-specific. Universally-harmed domains correspond to the largest calibration sharpening; an over-confident primary cannot integrate the helper’s signal because its prior precision dominates the likelihood update. *Empirical support:* not yet measured directly. Adjacent evidence: the 4B LoRA $r=128$ C2W/W2C ratio (19x on medicine, 1.4-2.0x on biology/law/math) is consistent with this hypothesis if calibration sharpening tracks domain identity. *Status:* the cleanest follow-up test we have not yet run; spinoff paper candidate (paper/spinoff_calibration.md).

5.3. Intruder dimensions (ruled out at the pairwise level)

Hypothesis: the “intruder dimensions” Shuttleworth et al. (2410.21228) document for LoRA-vs-full-FT differ in magnitude or density across domains, and the universally-harmed primaries accumulate more of them. *Empirical support:* the streaming weight-space CKA on the 10 LoRA adapters (§4.6) gives $r = -0.055$ between pairwise weight-space similarity and collab delta. Specialists are uniformly impaired in weight space rather than differentially. *Verdict:* the pairwise version of this hypothesis is ruled out. Per-adapter singular-value spectra could still differ; we have not measured that.

5.4. Bilinear rank constraint (ruled out at the rank levels we tested)

Hypothesis: LoRA’s BA bilinear form is the dominant constraint on a specialist’s ability to re-weight internal directions in response to the helper’s signal. *Empirical support:* our 4B LoRA $r=128$ data shows biology, law, and math specialists collaborating healthily at the same rank where medicine and physics collapse. The bilinear form is identical across all five; only the domain differs. *Verdict:* rank constraint at $r=128$ is not the dominant variable in our data. The role of LoRA’s parameterization at *much* higher rank ($r=512$, $r=1024$) remains an open follow-up question; preliminary $r=256$ / $r=512$ data is in progress.

5.5. Summary of mechanism status

The strongest empirical signal we have is *primary’s training-token entropy* (§5.1, $r=-0.380$), and it explains only $\sim 14\%$ of the variance. *Calibration sharpening* (§5.2) is the cleanest candidate to test next. Both *intruder dimensions* (§5.3) and *bilinear rank constraint* (§5.4) have been weakened by our data. Pinpointing the mechanism is left to follow-up work; the contribution of this paper is the asymmetry phenomenon and the negative results above.

6. Discussion

6.1. The cheap-deployment regime our claim covers

Our claim is scoped to a specific deployment regime: small open-source LLMs (Qwen3 1.7B and 4B in this paper) with cheap LoRA adapters as domain specialists, deliberating in pairs via alternating chain-of-thought. The asymmetry we find is large and reproducible in this regime. It is not a claim about flagship-model serial chains (Einstein Arena, Mixture-of-Agents over Claude / GPT-5 / Gemini) where participant capability is much higher, calibration deteriorates less post-fine-tuning, and the protocol is cumulative rather than alternating. We expect the failure modes we report to attenuate in that flagship regime; verifying that is open empirical work.

6.2. Implications for partner-selection heuristics

Today’s multi-agent system literature optimizes the *helper* slot: which model to call, what prompt template to use, how many rounds. Our data says the *primary* slot dominates the variance — 22x (CI [10x, 48x]) more than the helper slot — at matched solo accuracy in our regime. Partner-selection heuristics that ignore primary-agent properties (domain identity, training method) are optimizing the wrong dimension. A practical heuristic for deployments that use LoRA-fine-tuned specialists is: (a) establish whether the primary’s domain falls in the universally-harmed cluster (medicine, chemistry, physics, biology in our 1.7B data; medicine and physics under LoRA $r=128$ in our 4B data), and (b) if so, route to a single-agent (primary alone) or to a different training-method specialist (full fine-tuned) rather than to a peer- deliberation protocol. Concretely, for the universally-harmed primaries under LoRA at matched solo accuracy, multi-agent deliberation produces a 19x C2W / W2C switch ratio that we would not characterize as beneficial under any reasonable utility function.

6.3. Why the helper-slot variance is so small

The helper-slot variance is small in absolute terms (0.5% partial r^2 , column means span ~ 12 pp vs row spread ~ 49 pp). Our reading: under alternating chain-of-thought, the helper provides token-level context that the primary’s decoder either integrates or locks against; it does not appear to provide a separable propose-and-vote contribution. A flagship cumulative chain (§2.2) could plausibly produce larger helper-slot variance because each agent’s contribution accumulates rather than is overwritten. We do not have data to confirm that prediction; it is a clean falsification target.

6.4. The 1.7B-to-4B identity flip

Medicine moves from -38 pp at 1.7B to +9.5 pp at 4B; the Spearman rho between row-mean rankings on the 5 shared domains is -0.30 ($n=5$, low power). This is the single most fragile aspect of our

story. Two interpretations are open. (a) Configuration-dependent labels: scale, training method, and protocol all interact with the primary specialist’s brittleness; the *asymmetry* is invariant in magnitude but the *labels* of vulnerable primaries shift. (b) The 1.7B asymmetry is specific to the n=20 per-pair noise floor and a more powerful test would soften the universally-harmed claim. The N=200 1.7B 10x10 grid currently running will discriminate (a) from (b). Until that data lands, we report the asymmetry as the contribution and the labels as configuration-dependent.

7. Limitations and Future Work

The 10-domain grid at 1.7B is n=20 per pair, so individual-pair estimates carry wide confidence intervals (mean CI half-width ~18 pp). The aggregate row-level and ANOVA findings are robust to this noise, but individual pair-level claims are not. 7B replication is pending on the same infra. We stop at r=512 in the LoRA rank sweep. A fuller mechanism study is left to follow-up work; we stub four independent spinoffs below.

7.1. Future work: post-fine-tuning calibration across domains

(*Spinoff - paper/spinoff_calibration.md*)

7.2. Future work: token-space divergence as mechanism

(*In-progress analysis, may become §5 when data lands, otherwise spinoff.*)

7.3. Future work: information-theoretic framework

(*paper/spinoff_it_bayesian.md*)

7.4. Future work: training-data composition as a design variable

(*paper/spinoff_training_data.md*)

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